

Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

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Цели и задачи работы

Процесс выполнения лабораторной работы

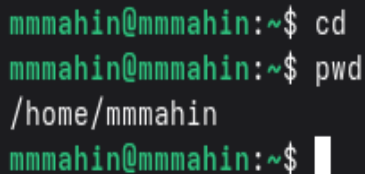
Выводы по проделанной работе

1. Цели и задачи работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду ls.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду history.

2. Процесс выполнения лабораторной работы



```
mmmahin@mmmahin:~$ cd
mmmahin@mmmahin:~$ pwd
/home/mmmahin
mmmahin@mmmahin:~$
```

A terminal window with a dark background. The prompt is 'mmmahin@mmmahin:~\$'. The user enters 'cd' and then 'pwd'. The output of 'pwd' is '/home/mmmahin'. The prompt is now 'mmmahin@mmmahin:~\$' with a white cursor block.

Рисунок 1: Путь к домашнему каталогу

Опции команды ls

```
mmmahin@mmmahin:~$ cd /tmp
mmmahin@mmmahin:/tmp$ ls
c8ed7dc6-489f-47d9-a983-193c63beec50.zip
happd.sock
hsperfdata_mmmahin
node-compile-cache
par-6d6d6d6168696e
snap-private-tmp
systemd-private-c4820e2599ff42be8f66b07162459b9f-abrt.service-I5ZRMQ
systemd-private-c4820e2599ff42be8f66b07162459b9f-chronyd.service-g7jrwG
systemd-private-c4820e2599ff42be8f66b07162459b9f-colord.service-J6PyrJ
systemd-private-c4820e2599ff42be8f66b07162459b9f-dbus-broker.service-3DzLuh
systemd-private-c4820e2599ff42be8f66b07162459b9f-irqbalance.service-FN7gFh
systemd-private-c4820e2599ff42be8f66b07162459b9f-low-memory-monitor.service-t3oPBu
systemd-private-c4820e2599ff42be8f66b07162459b9f-ModemManager.service-Xcl0a8
systemd-private-c4820e2599ff42be8f66b07162459b9f-pcsd.service-xT2r3R
systemd-private-c4820e2599ff42be8f66b07162459b9f-polkit.service-RRuQC1
systemd-private-c4820e2599ff42be8f66b07162459b9f-rtkit-daemon.service-JciQSz
systemd-private-c4820e2599ff42be8f66b07162459b9f-switcheroo-control.service-mpCEf1
systemd-private-c4820e2599ff42be8f66b07162459b9f-systemd-logind.service-BCgbUa
systemd-private-c4820e2599ff42be8f66b07162459b9f-systemd-resolved.service-ctJQsx
systemd-private-c4820e2599ff42be8f66b07162459b9f-upower.service-JfdPch
VMwareDnD
vmware-root_896-2730693439
mmmahin@mmmahin:/tmp$
```

Рисунок 2: Команда ls

Опции команды ls

```
mmmahin@mmmahin: /tmp$  
mmmahin@mmmahin: /tmp$ ls -a  
.  
..  
c8ed7dc6-489f-47d9-a983-193c63beec50.zip  
.font-unix  
happd.sock  
hsperfdata_mmmahin  
.ICE-unix  
node-compile-cache  
par-6d6d6d6168696e  
snap-private-tmp  
systemd-private-c4820e2599ff42be8f66b07162459b9f-abrt.service-I5ZRMQ  
systemd-private-c4820e2599ff42be8f66b07162459b9f-chronyd.service-g7jrwG  
systemd-private-c4820e2599ff42be8f66b07162459b9f-color.service-J6PyrJ  
systemd-private-c4820e2599ff42be8f66b07162459b9f-dbus-broker.service-3DzLuh  
systemd-private-c4820e2599ff42be8f66b07162459b9f-irqbalance.service-FN7gFh  
systemd-private-c4820e2599ff42be8f66b07162459b9f-low-memory-monitor.service-t3oPBu  
systemd-private-c4820e2599ff42be8f66b07162459b9f-ModemManager.service-Xcl0a8  
systemd-private-c4820e2599ff42be8f66b07162459b9f-pcsd.service-xT2r3R  
systemd-private-c4820e2599ff42be8f66b07162459b9f-polkit.service-RRuQC1  
systemd-private-c4820e2599ff42be8f66b07162459b9f-rtkit-daemon.service-JciQSz  
systemd-private-c4820e2599ff42be8f66b07162459b9f-switcheroo-control.service-mpCEFi  
systemd-private-c4820e2599ff42be8f66b07162459b9f-systemd-logind.service-BCgbUa  
systemd-private-c4820e2599ff42be8f66b07162459b9f-systemd-resolved.service-ctJQsx  
systemd-private-c4820e2599ff42be8f66b07162459b9f-upower.service-JfdPch  
VMwareDnD  
vmware-root_896-2730693439  
.X0-lock  
.X11-unix  
.X1-lock  
.XIM-unix  
mmmahin@mmmahin: /tmp$
```

Рисунок 3: Команда ls -a

```
mmmahin@mmmahin:/tmp$ ls -l
total 3872
-rw-r--r--. 1 mmahin mmahin 3961995 Sep  3 11:46 c8ed7dc6-489f-47d9-a983-193c63beec50.zip
srw-rw-rw-. 1 root   root      0 Sep  3 11:45 happd.sock
drwxr-xr-x. 2 mmahin mmahin    40 Sep  3 12:13 hsperfdata_mmmahin
drwxr-xr-x. 3 mmahin mmahin    60 Sep  3 12:23 node-compile-cache
drwx-----. 3 mmahin mmahin    60 Sep  3 12:12 par-6d6d6d6168696e
drwx-----. 2 root   root      40 Sep  3 11:44 snap-private-tmp
drwx-----. 3 root   root      60 Sep  3 11:44 systemd-private-c4820e2599ff42be8f66b07162459
b9f-abrtd.service-I5ZRMQ
drwx-----. 3 root   root      60 Sep  3 11:44 systemd-private-c4820e2599ff42be8f66b07162459
b9f-chronyd.service-g7jrw6
drwx-----. 3 root   root      60 Sep  3 11:45 systemd-private-c4820e2599ff42be8f66b07162459
b9f-colord.service-J6PyrJ
drwx-----. 3 root   root      60 Sep  3 11:44 systemd-private-c4820e2599ff42be8f66b07162459
b9f-dbus-broker.service-3DzLuh
drwx-----. 3 root   root      60 Sep  3 11:44 systemd-private-c4820e2599ff42be8f66b07162459
b9f-irqbalance.service-FN7gFh
drwx-----. 3 root   root      60 Sep  3 11:44 systemd-private-c4820e2599ff42be8f66b07162459
b9f-low-memory-monitor.service-t3oPBu
drwx-----. 3 root   root      60 Sep  3 11:44 systemd-private-c4820e2599ff42be8f66b07162459
b9f-ModemManager.service-Xcl0a8
drwx-----. 3 root   root      60 Sep  3 11:45 systemd-private-c4820e2599ff42be8f66b07162459
b9f-pcsd.service-xT2r3R
drwx-----. 3 root   root      60 Sep  3 11:44 systemd-private-c4820e2599ff42be8f66b07162459
b9f-polkit.service-RRuQCl
drwx-----. 3 root   root      60 Sep  3 11:44 systemd-private-c4820e2599ff42be8f66b07162459
b9f-rtkit-daemon.service-lci0Sz
```

Рисунок 4: Команда ls -l

Опции команды ls

```
mmmahin@mmmahin:/tmp$ ls -f
..
node-compile-cache
hsperfdata_mmmahin
par-6d6d6d6168696e
VMwareDnD
c8ed7dc6-489f-47d9-a983-193c63beec50.zip
.X1-lock
.X0-lock
systemd-private-c4820e2599ff42be8f66b07162459b9f-pcsd.service-xT2r3R
systemd-private-c4820e2599ff42be8f66b07162459b9f-colord.service-J6PyrJ
happd.sock
systemd-private-c4820e2599ff42be8f66b07162459b9f-ModemManager.service-Xcl0a8
vmware-root_896-2730693439
systemd-private-c4820e2599ff42be8f66b07162459b9f-systemd-logind.service-BCgbUa
systemd-private-c4820e2599ff42be8f66b07162459b9f-upower.service-JfdPch
systemd-private-c4820e2599ff42be8f66b07162459b9f-switcheroo-control.service-mpCEFi
systemd-private-c4820e2599ff42be8f66b07162459b9f-rtkit-daemon.service-JciQSz
systemd-private-c4820e2599ff42be8f66b07162459b9f-polkit.service-RRuQCl
systemd-private-c4820e2599ff42be8f66b07162459b9f-low-memory-monitor.service-t3oPBu
systemd-private-c4820e2599ff42be8f66b07162459b9f-chrond.service-g7jrw6
systemd-private-c4820e2599ff42be8f66b07162459b9f-irqbalance.service-FN7gFh
systemd-private-c4820e2599ff42be8f66b07162459b9f-abrt.service-I5ZRMQ
systemd-private-c4820e2599ff42be8f66b07162459b9f-dbus-broker.service-3DzLuh
systemd-private-c4820e2599ff42be8f66b07162459b9f-systemd-resolved.service-ctJQsx
.font-unix
.XIM-unix
.ICE-unix
.X11-unix
snap-private-tmp
mmmahin@mmmahin:/tmp$
```

Рисунок 5: Команда ls -f

```
mmmahin@mmmahin:/tmp$  
mmmahin@mmmahin:/tmp$ cd /var/spool/  
mmmahin@mmmahin:/var/spool$ ls -l  
total 0  
drwxr-x--x. 1 root abrt 24476 Aug 31 13:37 abrt  
drwx-----. 1 abrt abrt    0 Dec  4 2025 abrt-upload  
drwxr-xr-x. 1 root root   66 Oct 23 2025 anacron  
drwx-----. 1 root root   18 Oct 23 2025 at  
drwx-----. 1 root root    0 Aug  6 2025 cron  
drwx--x---. 1 root lp     6 Dec 12 2025 cups  
drwxr-xr-x. 1 root root    0 Jul 30 2025 lpd  
drwxrwxr-x. 1 root mail 258 Sep  3 11:44 mail  
drwxr-xr-x. 1 root root    0 Jan 27 2026 plymouth  
mmmahin@mmmahin:/var/spool$
```

Рисунок 6: Каталог /var/spool

```
mmmahin@mmmahin:/var/spool$ cd
mmmahin@mmmahin:~$ ls
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work
mmmahin@mmmahin:~$ ls -al
total 24
drwx-----. 1 mmmahin mmmahin 370 Sep  3 12:26 .
drwxr-xr-x. 1 root    root    258 Sep  3 11:44 ..
-rw-----. 1 mmmahin mmmahin 1420 Sep  3 12:40 .bash_history
-rw-r--r--. 1 mmmahin mmmahin  18 Jul 23  2025 .bash_logout
-rw-r--r--. 1 mmmahin mmmahin 144 Jul 23  2025 .bash_profile
-rw-r--r--. 1 mmmahin mmmahin 677 Sep  3 12:23 .bashrc
drwx-----. 1 mmmahin mmmahin 484 Sep  3 12:28 .cache
drwx-----. 1 mmmahin mmmahin 360 Sep  3 12:40 .config
drwxr-xr-x. 1 mmmahin mmmahin   0 Sep  3 11:46 Desktop
drwxr-xr-x. 1 mmmahin mmmahin   0 Sep  3 11:46 Documents
drwxr-xr-x. 1 mmmahin mmmahin   0 Sep  3 11:46 Downloads
-rw-r--r--. 1 mmmahin mmmahin 334 Nov 14  2025 .emacs
-rw-r--r--. 1 mmmahin mmmahin 233 Sep  3 12:05 .gitconfig
drwxr-xr-x. 1 mmmahin mmmahin  74 Sep  3 12:37 git-extended
drwx-----. 1 mmmahin mmmahin 136 Sep  3 12:04 .gnupg
drwx-----. 1 mmmahin mmmahin  20 Sep  3 11:46 .local
drwxr-xr-x. 1 mmmahin mmmahin  34 Oct 23  2025 .mozilla
drwxr-xr-x. 1 mmmahin mmmahin   0 Sep  3 11:46 Music
drwxr-xr-x. 1 mmmahin mmmahin   0 Sep  3 11:46 Pictures
drwxr-xr-x. 1 mmmahin mmmahin   0 Sep  3 11:46 Public
drwx-----. 1 mmmahin mmmahin 132 Sep  3 12:06 .ssh
drwxr-xr-x. 1 mmmahin mmmahin   0 Sep  3 11:46 Templates
drwxr-xr-x. 1 mmmahin mmmahin  18 Sep  3 12:11 .texlive2025
drwxr-xr-x. 1 mmmahin mmmahin   0 Sep  3 11:46 Videos
drwxr-xr-x. 1 mmmahin mmmahin  10 Sep  3 12:06 work
mmmahin@mmmahin:~$
```

Рисунок 7: Файлы в домашнем каталоге

```
mmmahin@mmmahin:~$ mkdir newdir
mmmahin@mmmahin:~$ mkdir newdir/morefun
mmmahin@mmmahin:~$ mkdir letters memos misk
mmmahin@mmmahin:~$ ls
Desktop  Downloads  letters  misk  newdir  Public  Videos
Documents  git-extended  memos  Music  Pictures  Templates  work
mmmahin@mmmahin:~$ rm letters/ memos/ misk/
rm: cannot remove 'letters/': Is a directory
rm: cannot remove 'memos/': Is a directory
rm: cannot remove 'misk/': Is a directory
mmmahin@mmmahin:~$ rm -r letters/ memos/ misk/
mmmahin@mmmahin:~$ rm -r newdir/
mmmahin@mmmahin:~$ ls
Desktop Documents Downloads git-extended Music Pictures Public Templates Videos work
mmmahin@mmmahin:~$
```

Рисунок 8: Действия с каталогами

```
'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/image':
solvay.jpg

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources':
csl tex

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/csl':
gost-r-7-0-5-2008-numeric.csl

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/tex':
preamble.tex

'./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':
image-report mpv-shot
mmmahin@mmmahin:~$ ls -t
git-extended work Desktop Documents Downloads Music Pictures Public Templates Videos
mmmahin@mmmahin:~$
```

Рисунок 9: Команда ls -R и ls -t

```
mmmahin@mmmahin:~$ help cd
cd: cd [-L|[-P [-e]]] [-@] [dir]
    Change the shell working directory.

Change the current directory to DIR.  The default DIR is the value of the
HOME shell variable.  If DIR is "-", it is converted to $OLDPWD.

The variable CDPATH defines the search path for the directory containing
DIR.  Alternative directory names in CDPATH are separated by a colon (:).
A null directory name is the same as the current directory.  If DIR begins
with a slash (/), then CDPATH is not used.

If the directory is not found, and the shell option `cdable_vars' is set,
the word is assumed to be a variable name.  If that variable has a value,
its value is used for DIR.

Options:
    -L      force symbolic links to be followed: resolve symbolic
            links in DIR after processing instances of `..'
    -P      use the physical directory structure without following
            symbolic links: resolve symbolic links in DIR before
            processing instances of `..'
    -e      if the -P option is supplied, and the current working
            directory cannot be determined successfully, exit with
            a non-zero status
    -@      on systems that support it, present a file with extended
            attributes as a directory containing the file attributes

The default is to follow symbolic links, as if `-L' were specified.
`..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.
```

Рисунок 10: Справка по команде cd

```
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the version de-
    scribed here. Please refer to your shell's documentation for details about the options
    it supports.

AUTHOR
    Written by Jim Meyering.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their file modes
        unaffected by any -m option

    -v, --verbose
        print a message for each created directory

    -Z
        set SELinux security context of each created directory to the default type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK security context to
        CTX

    --help display this help and exit

    --version
        output version information and exit

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir

```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b
        a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir

```
RM(1)                                User Commands                                RM(1)

NAME
    rm - remove files or directories

SYNOPSIS
    rm [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of rm.  rm removes each specified file.  By
    default, it does not remove directories.

    If the -I or --interactive=once option is given, and there are more than three files or
    the -r, -R, or --recursive are given, then rm prompts the user for whether to proceed
    with the entire operation.  If the response is not affirmative, the entire command is
    aborted.

    Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force
    option is not given, or the -i or --interactive=always option is given, rm prompts the
    user for whether to remove the file.  If the response is not affirmative, the file is
    skipped.

OPTIONS
    Remove (unlink) the FILE(s).

    -f, --force
        ignore nonexistent files and arguments, never prompt

    -i
        prompt before every removal

    -I
        prompt once before removing more than three files, or when removing recursively;
        less intrusive than -i, while still giving protection against most mistakes

Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm

```
100  cd /var/spool/
101  ls -l
102  cd
103  ls
104  ls -al
105  mkdir newdir
106  mkdir newdir/morefun
107  mkdir letters memos misk
108  ls
109  rm letters/ memos/ misk/
110  rm -r letters/ memos/ misk/
111  rm -r newdir/
112  ls
113  ls -R
114  ls -t
115  help cd
116  man pwd
117  man mkdir
118  man rmdir
119  man rm
120  history
mmmahin@mmmahin:~$
```

Рисунок 15: Команда history

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.